MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: PL Roll No.: 27 Exp. No.:

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Performed on:

Submitted on:

Marks	Teacher's Signature with date
8	
10	

Q.1. List the python sequence data types and state one key characteristic of each with respect to mutability of ordering

→ In python, sequence data types store collections of items in a specific order.

1. List :→ Ordered sequence
   → Mutable [elements can be changed after creation]
2. Tuple :→ Ordered sequence
   → Immutable [cannot be created once created]
3. String :→ Ordered sequence of characters
   → Immutable [characters cannot be modified]
4. Range :→ Ordered sequence of numbers
   → Immutable [cannot modify values, only generate them]

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

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Q.2. Explain how indexing and slicing work in python sequences. Illustrate how they differ when applied to strings and lists.

→ Indexing in python sequences.

Indexing is used to access a single element from a sequence.

- Indexing starts from 0 [Left to Right]
- Negative indexing starts from -1 [Right to left]

Ex:- code

```
s = "python"
l = [10, 20, 30, 40]
print(s[0])
print(s[-1])
print(l[1])
print(l[-2])
```

- Slicing in python sequences.

Slicing is used to extract a part (sub sequence) of a sequence.

- Syntax :

sequence[start : stop : step]

start ⟶ starting index [included]
stop ⟶ ending index [excluded]
step ⟶ gap between elements [optional]

Ex:

```
print(s[1:4])
print(l[0:3])
print(s[: : 2])
print(l[: : -1])
```

Q.3. Write a python program that :

- Uses indexing and slicing on a string List, and.
- Applies at least two built-in function to display meaningful result.

→ python code :-

```
text = "python programming"
# indexing
first_char = text[0]
last_char = text[-1]

# slicing
sub_text = text[0:6]
every_second = text[::2]

length = len(text)
max_char = max(text)

print("original string :", text)
print("first character :" first_char)
print("last character :" last_char)
print("sliced string (0:6) :", sub_text)
print("Every second character :", every_second)
print("length of string :", length)
print("maximum character :" max_char)
```

- indexing → Accessing single characters (text[0], text[-])
- slicing → Extracting part of the string (text[0:6], text[::2])
- Built in functions used
  - len() → Finds length of the string
  - max() → Finds the highest characters (based on ASCII value)

Q.4. Analyze the difference between equality (==) and identity (is) when used with sequence types. Discuss how object mutability and memory reference affect program behaviour.

→ Difference between == and is in python.

- The equality operator (==) compares the values or contents of two sequence objects.
- The identity operator (is) compares the memory location of two objects.
- Two sequences may have the same values but be stored in different memory locations. so == returns true while is returns false.
- For mutable sequence like lists, if two variable refer to the same object, changes made through one variable affect the other.
- For immutable sequence like string and tuples, values cannot be modified, any change creates a new object, reducing side effects.

Q.5. Evaluate the suitability of lists, sets, and dictionaries for storing and managing unique student records. Justify which structure is most efficient and why.

→ Suitability of lists, sets and dictionaries for unique student records.

When storing and managing unique student records, the choice of dates structure affects efficiency, Integrity and ease of access.

1 Lists

- Lists store elements in an ordered sequence.
- They allow duplicate entries, so extra logic is required to ensure uniqueness.
- Searching for a student record is slow especially for large datasets.

2 Sets.

- Sets store only unique elements automatically
- They are unordered and do not support indexing.
- Fast membership checking (O(1) average (time).

3 Dictionaries.

- It store data as key - value pairs.
- Keys are unique, ensuring no duplicate student IDS.
- Very fast access, insertion and deletion (O(1) average time)
- Can store structured records as values.

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Q.6. Design and implement a python program that
(1) Appropriate sequence types
(2) Indexing or slicing is built in functions to manage and process student data efficiency.

→

```
students = {
1 : {"name" : "Amit", "marks" : [78, 85, 90]}
2 : {"name" : "Sneha", "marks" : [88, 91, 81]}

for roll, data in students.items():
    marks = data["marks"]

first_mark = mark[0]
top_two = sorted(marks, reverse = true)[:2]

total = sum(marks)
average = total / len(marks)

print(F"\n Roll no : {roll}")
print(F"name : {data['name']}")
print(F"marks : {marks}")
print(F"First mark : {first_mark}")
```

Explaination :
- Dictionary ensures unique student
- List store marks
- Indexing (marks[0] and slicing [:2]) are used.
- Buit in function : sum(), len(), sorted()

Q.7. Write a program to print the first and last character of a string.

```
text = input ("Enter a string :")

print ("First character :", text [0])
print (" Last character : ", text [-1])
```

- text [0] → access the first character.
- text [-1] → access the last character using negative indexing.

Q.8. Write a program to create a list, tuple, set and dictionary

```
My_List = [10, 20, 30, 40]

My_tuple = (1, 2, 3, 4)

My_set = {5, 10, 15}

My_dict = {5, 10, 15}

My_dict = {
          "name": "Ampt",
          "age" : 20,
          "branch": "CSF"
          }
```

```
print ("List :", my_list)
print ("Tuple :", my_tuple)
print ("set :", my_set)
print ("Dictionary :", my_dict)
```

Q.9. Write a program to understand the concept of indexing.

→

```
text = "python"

print ("string :", text)
print ("first character :", text[0])
print ("second character :", text[1])
print ("last character :", text[-1])

numbers = [10, 20, 30, 40]
print ("In List :", numbers)
```

Q.10. Write a program to understand the concept of built-in function.

→

```
Numbers = [10, 20, 30, 40, 50]
print ("List :", numbers)

print ("length of list :", len(numbers)))
print ("maximum value:", max(numbers))
print ("minimum value :", min(numbers))
print ("sum of elements :", sum(numbers))
print ("sorted List :" sorted (numbers))
```